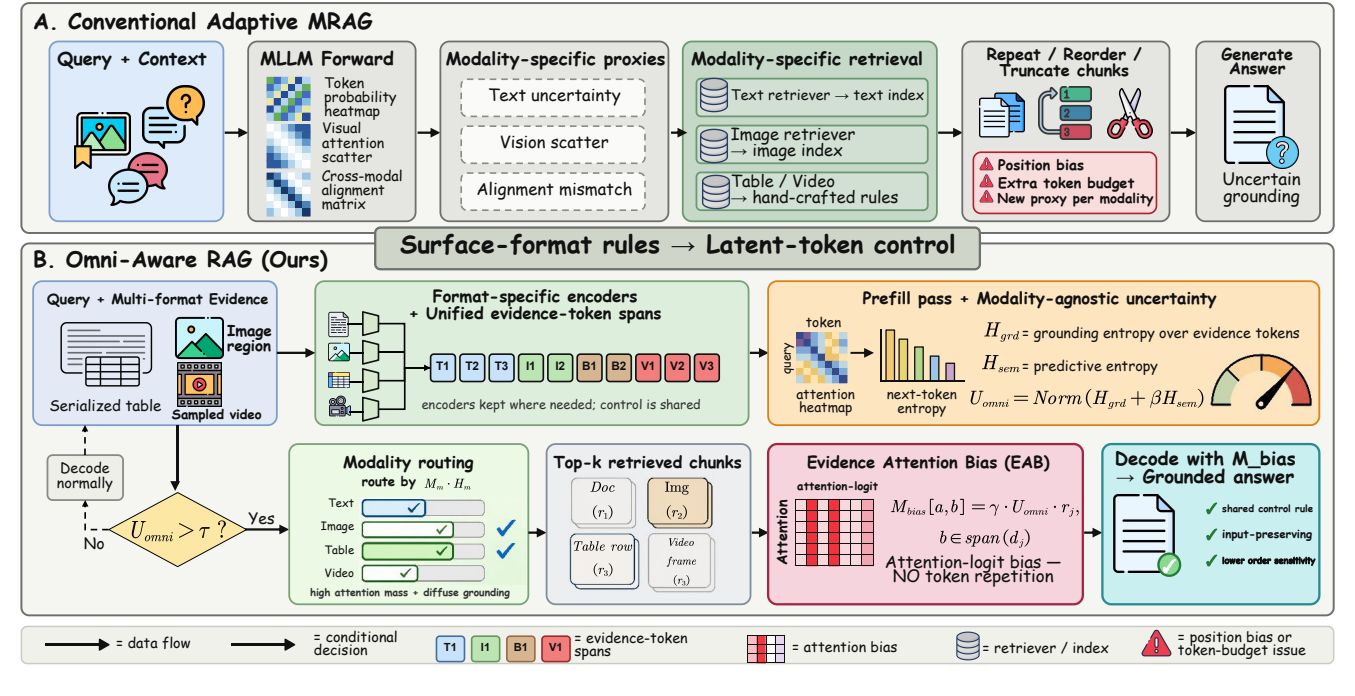

A. Conventional Adaptive MRAG
Query + Context
MLLM Forward
Token probability heatmap
Visual attention scatter
Cross-modal alignment matrix
Modality-specific proxies
Text uncertainty
Vision scatter
Alignment mismatch
Modality-specific retrieval
Text retriever → text index
Image retriever → image index
Table / Video → hand-crafted rules
Repeat / Reorder / Truncate chunks
Position bias
Extra token budget
New proxy per modality
Generate Answer
Uncertain grounding
Surface-format rules → Latent-token control
B. Omni-Aware RAG (Ours)
Query + Multi-format Evidence
Image region
Serialized table
Sampled video
Format-specific encoders + Unified evidence-token spans
T1 T2 T3 I1 I2 B1 B2 V1 V2 V3
encoders kept where needed; control is shared
Prefill pass + Modality-agnostic uncertainty
token
query
attention heatmap
next-token entropy
H_{grd} = grounding entropy over evidence tokens
H_{sem} = predictive entropy
$U_{omni} = Norm(H_{grd} + \beta H_{sem})$
Decode normally
$U_{omni} > \tau$?
Yes
No
Modality routing
route by $M_m \cdot H_m$
Text
Image
Table
Video
high attention mass + diffuse grounding
Top-k retrieved chunks
Doc (r_1)
Img (r_2)
Table row (r_3)
Video frame (r_3)
Evidence Attention Bias (EAB)
attention-logit
Attention
$M_{bias}[a,b] = \gamma \cdot U_{omni} \cdot r_j,$
$b \in span(d_j)$
Attention-logit bias — NO token repetition
Decode with M_bias → Grounded answer
shared control rule
input-preserving
lower order sensitivity
= data flow
= conditional decision
T1 I1 B1 V1 = evidence-token spans
= attention bias
= retriever / index
= position bias or token-budget issue